TITLE: TECHNICAL CHALLENGE INTERN PROGRAM – 2026 – Communication to Applicants -

Email message to send to each applicant-

"¡Hola!

Queremos agradecerte por tu interés en ser parte del programa de pasantías en Proofpoint – 2026. Esperamos que el proceso de selección sea una gran experiencia y un desafío enriquecedor.

Antes de avanzar a la etapa de entrevistas, te enviamos algunas preguntas y unos ejercicios prácticos que deberás completar y enviar antes del JUEVES 12/03 *al final del día* por este medio al siguiente mail: University Recruiting UniversityRecruiting@proofpoint.com

A-Por favor, responder a las siguientes preguntas lo más preciso posible, preferentemente en Inglés.

1. Describe the most relevant learnings from programming-related subjects during your career.
2. Describe a situation you had to resolve with a teammate from a study or work group.
3. Describe your personal learning plans or development expectations in technical areas in the short and long term.

B- Ejercicio Práctico a resolver:

The Streaming Service's Lost Episodes

Context

A streaming platform is digitizing its catalog of series, seasons, and episodes.
However, during the ingestion of data, no validation or uniqueness checks were applied, which resulted in a corrupted catalog containing missing information, inconsistent values, and duplicated episodes.

You will be given a CSV file where each row represents an episode.
Each entry should contain the following fields in this order:

- Series Name – name of the series (required) str
- Season Number – season number (may be missing or invalid) int
- Episode Number – episode number (may be missing or invalid) int
- Episode Title – episode title (may be missing) str
- Air Date – date when the episode aired (may be missing or invalid) str

Because of the lack of checks during data entry, the file may contain inconsistencies, incorrect formats, and repeated entries.

Goal

Your task is to process this CSV file and produce a clean, corrected, and de-duplicated catalog.
You must:

- Parse and normalize all entries
- Correct missing or invalid information following the rules below
- Detect and remove duplicated episodes
- Keep only one valid record for each unique episode
- Produce a final, well-organized list of episodes
- Generate a quality report summarizing your corrections

Additional organization (e.g., grouping or sorting) is a plus but not mandatory.

Problems You May Encounter in the File

1. **Missing or Empty Fields**

   The following fields may be empty or missing entirely:

- Series Name
  If missing or empty → discard the record (required to identify an episode)
- Season Number
  If missing, empty, negative, or not a number → set to 0
- Episode Number
  If missing, empty, negative, or not a number → set to 0
- Episode Title
  If missing or empty → replace with "Untitled Episode"
- Air Date
  If missing, empty, or invalid → replace with "Unknown"
- When Episode Number, Episode Title **and** Air Date are missing (the three fields), discard the record (required to identify and merge)
  Consider values 0 and "Unknown" as missing information

2. **Invalid Data Formats**

   You may find:

- Non-numeric values in Season or Episode numbers
  e.g., "one", "3.5", "--2"
- Impossible dates
  e.g., "not a date", "2022-40-99", "0000-00-00"
- Extra whitespace, inconsistent capitalization, strange characters, etc.

All such cases should be cleaned and normalized.

3. **Duplicate Entries**

   Because no uniqueness constraints were enforced during ingestion, the same episode may appear:

- several times,
- with different titles,
- different air dates,
- different spacing or formatting,
- or inconsistent season/episode numbering.

Episodes must be considered duplicates when they refer to the same:

```
(SeriesName_normalized, SeasonNumber, EpisodeNumber)
```

Or

```
(SeriesName_normalized, 0, EpisodeNumber, EpisodeTitle_normalized)
```

Or

```
(SeriesName_normalized, SeasonNumber, 0, EpisodeTitle_normalized)
```

Normalized means: trimmed, collapsed spaces, and lowercased for comparison.

When duplicates exist, you must keep only the “best” record, following this priority:

- Episodes with a valid Air Date over "Unknown"
- Episodes with a known Episode Title over "Untitled Episode"
- Episodes with a valid Season Number and Episode Number
- If still tied → keep the first entry encountered in the file

Output

You must provide the source code (any language of your preference) of a program that can read the csv input file. Feel free to share a Github link if that's easier. The code should generate the following output:

1. **Cleaned Episode Catalog**

   Output file name: **episodes_clean.csv**

   With columns:

   `SeriesName,SeasonNumber,EpisodeNumber,EpisodeTitle,AirDate`

   Requirements:

   - All corrections applied
   - No duplicates
   - Safe defaults used where appropriate
   - Each row represents a valid episode

2. **Data Quality Report**

   Output file name: **report.md**

   Must include:

   - Total number of input records
   - Total number of output records
   - Number of discarded entries
   - Number of corrected entries
   - Number of duplicates detected
   - Explanation of the deduplication strategy

   This report helps assess data quality and your decision-making

C- Adicionalmente, si alguien cuenta con tiempo, les enviamos este otro ejercicio:

Análisis de Frecuencia de Palabras

Escribe un programa que lea un archivo de texto y realice un análisis de frecuencia de palabras:

- El programa debe ser capaz de leer un archivo de texto proporcionado como entrada y contar cuántas veces aparece cada palabra en el texto.
- La comparación de palabras debe ser sin distinción entre mayúsculas y minúsculas y no considerar signos de puntuación y caracteres especiales.
- Mostrar las 10 palabras más frecuentes, junto con su frecuencia de aparición.

Fecha límite de entrega: JUEVES 12 de MARZO (final del día) respondiendo a este mail.

Si necesitas consultar alguno de los puntos, no dudes en enviar un mensaje por mail y te ayudaremos a resolver la consulta.

Muchas gracias y quedamos a disposición por cualquier consulta.

University Relations Team”